V4 Phase 2b charger / power stage — review checkpoint

Input power connector

J1
ADAPTER 4.75–21V
PWR_FLAG
F1
0451005.MRL 5A
VIN1
PGND

Battery connector

J3
BATTERY 2S/4S
BATTERY
PGND

Output power connector

J2
VSYS OUTPUT
SYS
PGND

F1 prototype: Littelfuse 0451005.MRL, 5 A, 125 Vdc; qualify temperature, cable, hot-plug, polarity-FET SOA and coordination.

TI Fig. 8-1 VIN damping: 2 ohm series plus 2.2 uF shunt to PGND; exact >=35 V parts pending procurement.

Q1
AON7528
Q2
AON7528
D1
12V
R1
270R
ACDRV1
VIN1
NT4
VBUS

IC1
BQ25798RQMR
VBUS
VIN1
ACDRV1
VBUS
PGND
D-
D+
I2C_SCL
I2C_SDA
-INT
CE_N
QON
ILIM_HIZ
PROG
TS
PMID
BTST1
SW1
SW2
BTST2
STAT
REGN
SYS
BAT
SDRV
GND
NT1
PGND

R4
100R
BATP
BATTERY
C23
47n 25V
R5
27k 1%
PROG
GND

TP3
PGND
TP4
SYS
TP1
GND

boost side
BTST2
47n 25V
C1
SW2
L1
2.2uH
buck side
SW1
47n 25V
C2
BTST1

-INT
CHG_REQ
I2C_SCL
I2C_SDA
MAIN_3V3
J4
MCU / BIAS INTERFACE
PGND

The REGN LDO supplies internal bias circuits and the MOSFETs gate drivers.
The pull-up rails of ILIM_HIZ, TS, and STAT can be connected to REGN.
The REGN is not recommended to source any other external circuit!

REGN
R8
2k2
LED
D2
STAT
C3
4.7u 25V
PGND
NT2
ILIM_HIZ
R6
5k6
R7
33k
TS
NTC temp sensor
J6
GND

Grounds
PGND
PWR_FLAG
GND
PWR_FLAG
PWR_FLAG
VBUS

VIN hot-plug RC: TI Fig. 8-1 (>15V)
Provisional R12 / C26; validate on prototype

PMID bulk C20/C22/C24/C25: DNP in initial VSYS profiles; backup bulk is deferred.

BATTERY
C10 10u 25V
C4 10u 25V
PGND
SYS
C21 10u 25V
C19 10u 25V
C17 10u 25V
C14 10u 25V
C11 10u 25V
C5 100n 25V
PMID
C24
C25
C22
C20 10u 50V
C18 10u 50V
C15 10u 50V
C12 10u 50V
C6 100n 50V
VBUS
C16 10u 50V
C13 10u 50V
C7 100n 50V

PGND = system return; GND = quiet TS / PROG / NTC; NT1 local tie; no PCB moat.

SDRV
C8
1nF, 50V

CE_N defaults OFF. MAIN_3V3 / CHG_REQ are the only intended control paths; USB 3V3 must not bias CE.

VIN1
C9
100n 50V

VIN1
R12
2R 5% pulse
C26
2.2u 100V X7R

REGN
BAS116H
D3
CE_BIAS
MAIN_3V3
BAS116H
D4
CE_BIAS
R9
10k
CE_N
R11
1k
CHG_REQ
CHG_GATE
Q3
PMV16UN
R10
10k

H1
H2
H3
H4

Sheet: /
File: ups12v_v4.kicad_sch

Title: BQ25798 UPS module

Size: A3	Date:	Rev: 2.1
KiCad E.D.A. 10.0.7		Id: 1/1